

5-Fold Cross Validation Evaluation Report (Ensemble + Calibration)

Started: 2026-04-14 14:44:36

Finished: 2026-04-14 15:13:18

Model Architecture: MobileNetV2

Phase 1 LR: 0.01 | Phase 2 LR: 1e-06

Aggregate Performance across 5 Folds (Individual Optimized Thresholds):

Average Accuracy: 0.8698 ± 0.0072

Average Sensitivity: 0.8784 ± 0.0528

Average Specificity: 0.8608 ± 0.0421

Average AUC: 0.9354 ± 0.0015

CALIBRATED ENSEMBLE TEST METRICS:

Forced Decision Threshold: 0.5000

Accuracy : 0.8405

Precision : 0.9134

Sensitivity (Recall) : 0.7582

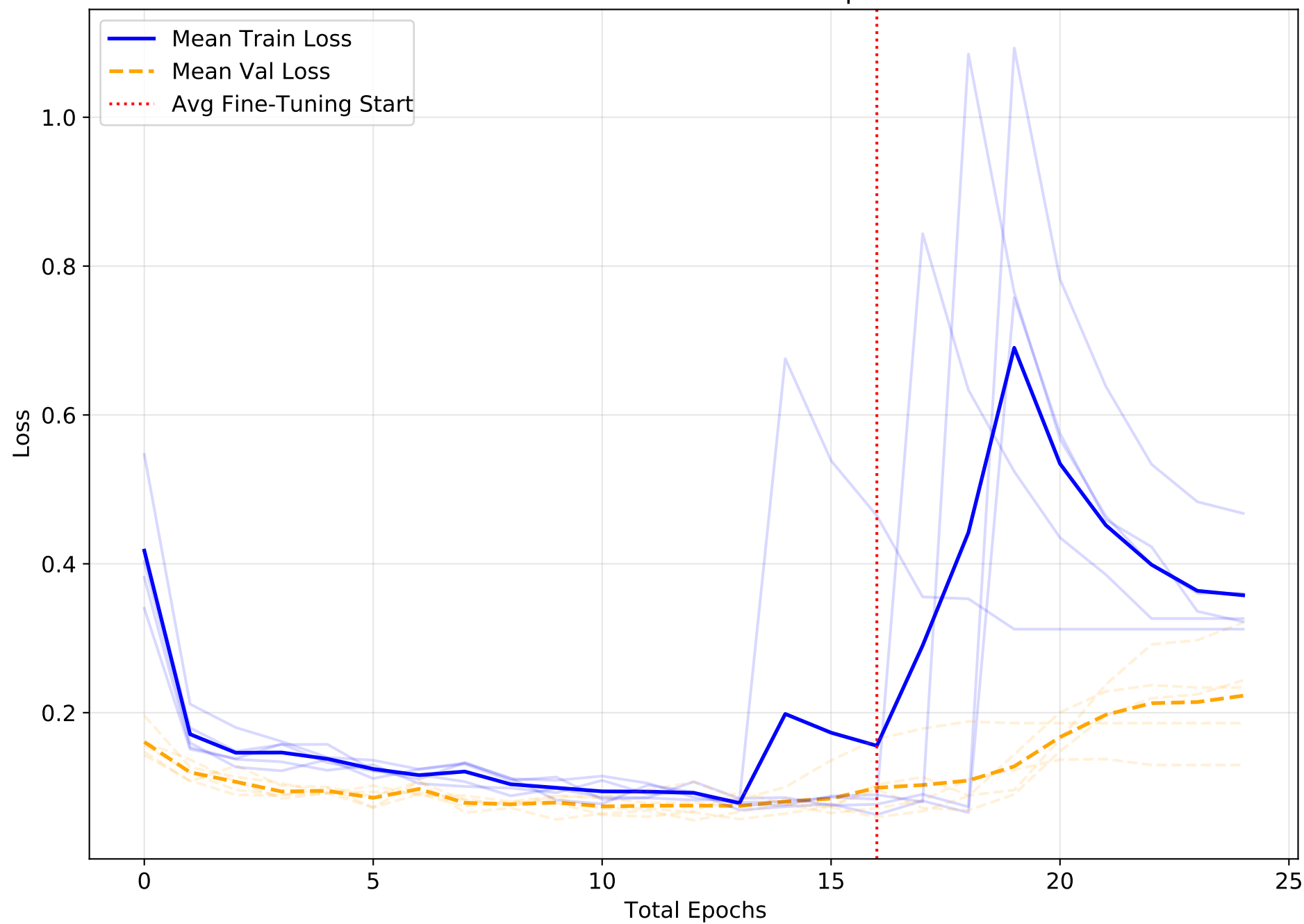
Specificity : 0.9257

Ensemble AUC : 0.9377

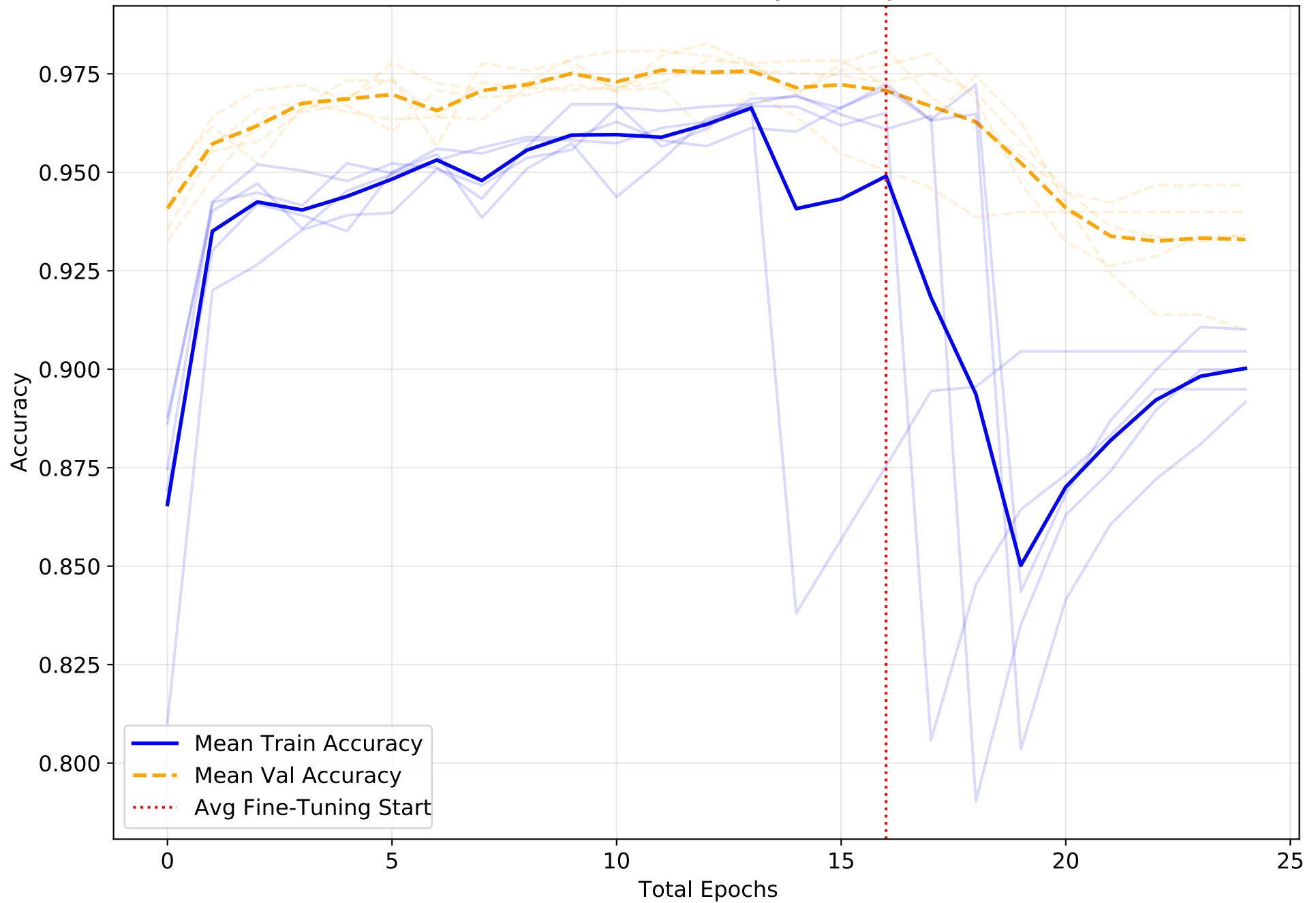
Classification Report (at threshold 0.5000):

	precision	recall	f1-score	support
Normal	0.79	0.93	0.85	296
Cataract	0.91	0.76	0.83	306
accuracy			0.84	602
macro avg	0.85	0.84	0.84	602
weighted avg	0.85	0.84	0.84	602

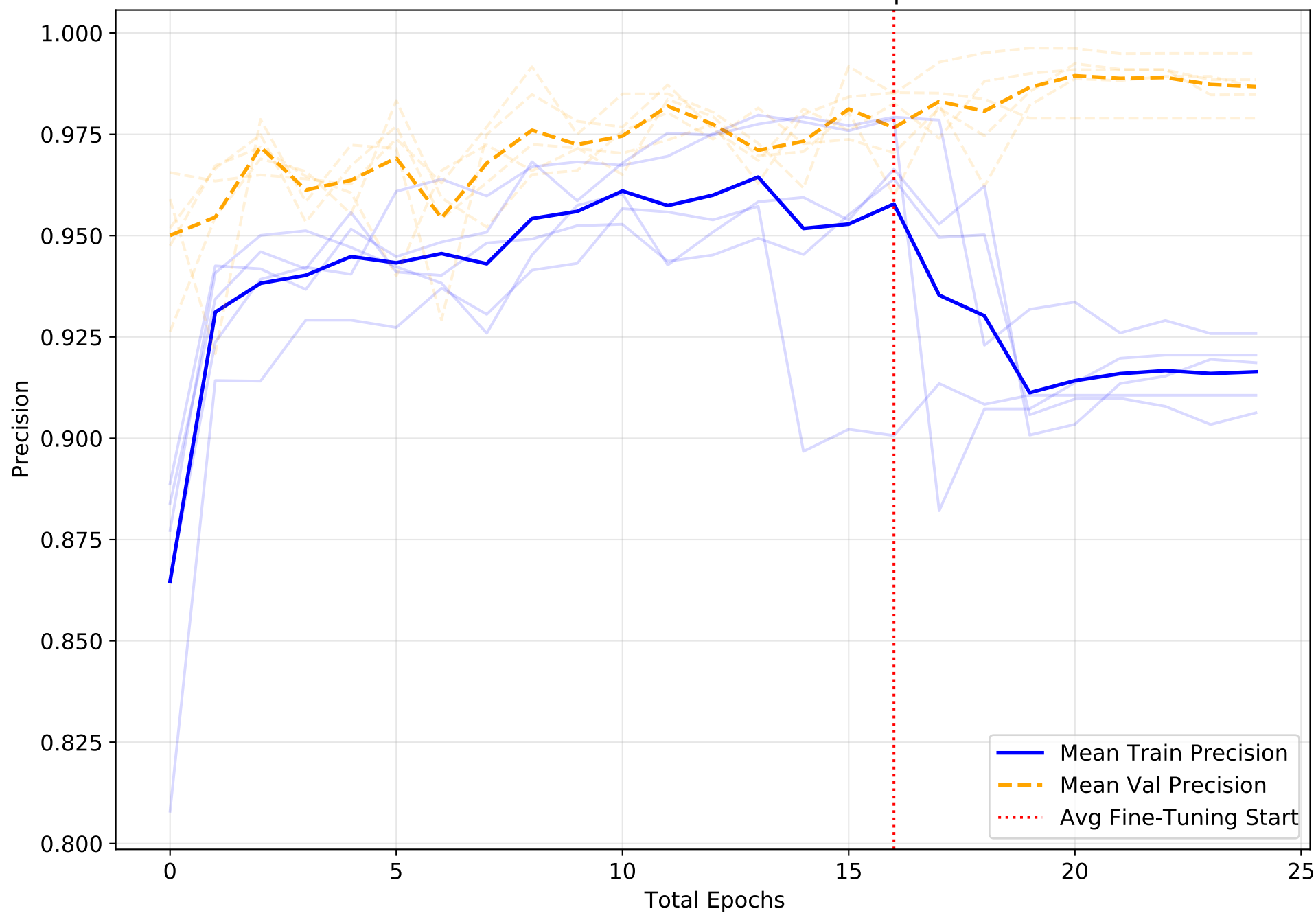
Cross-Validated Loss Over Epochs



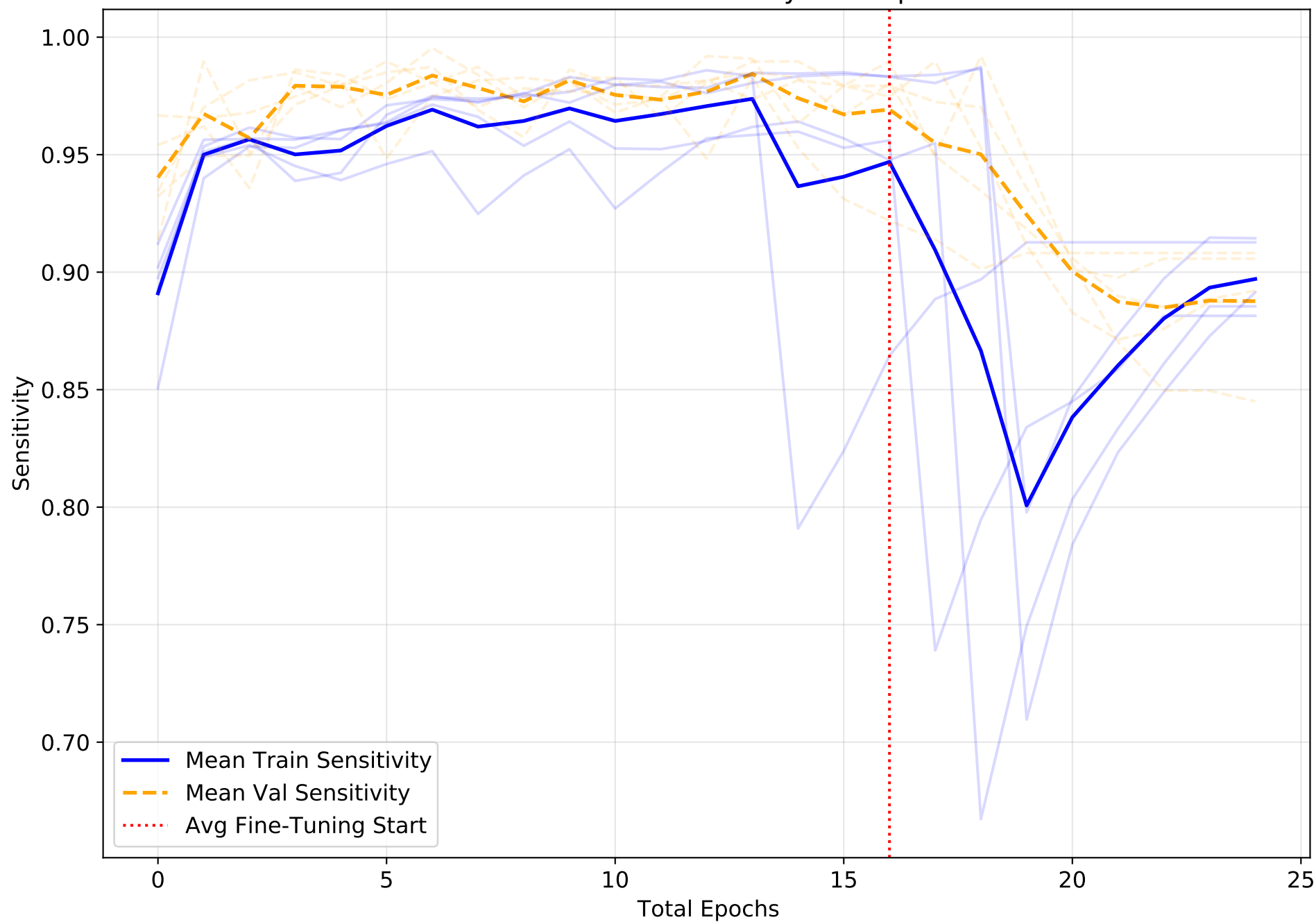
Cross-Validated Accuracy Over Epochs



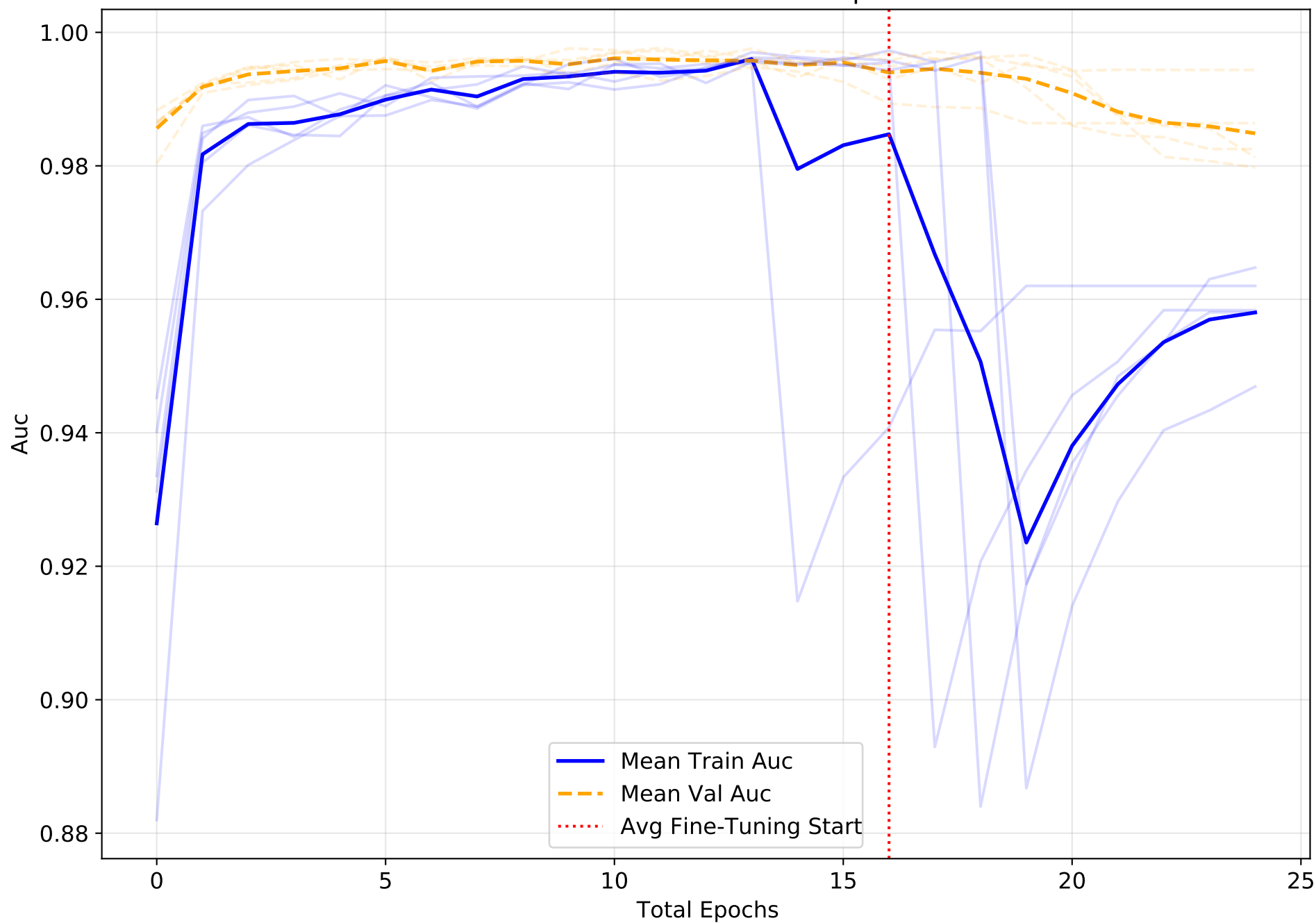
Cross-Validated Precision Over Epochs



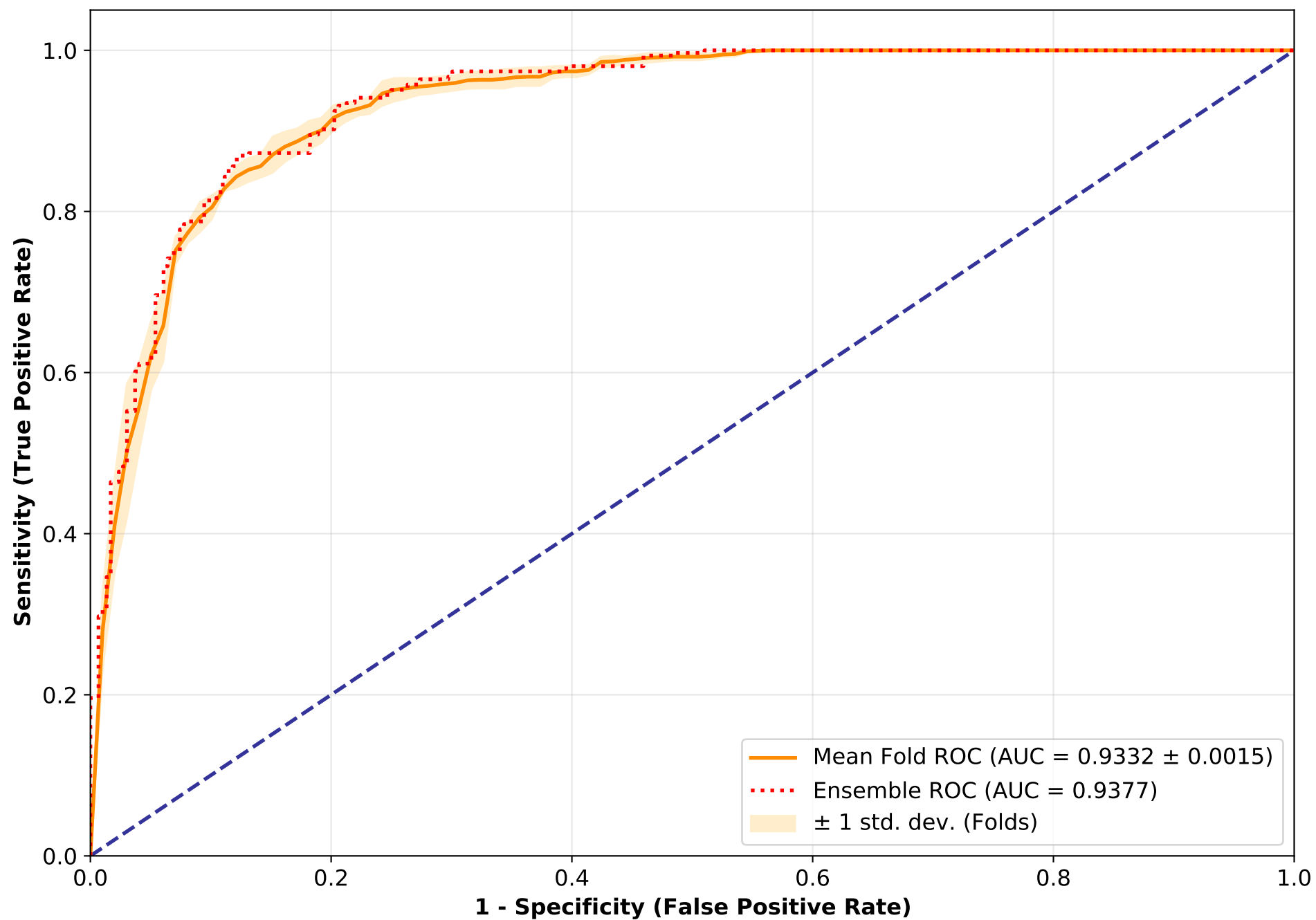
Cross-Validated Sensitivity Over Epochs



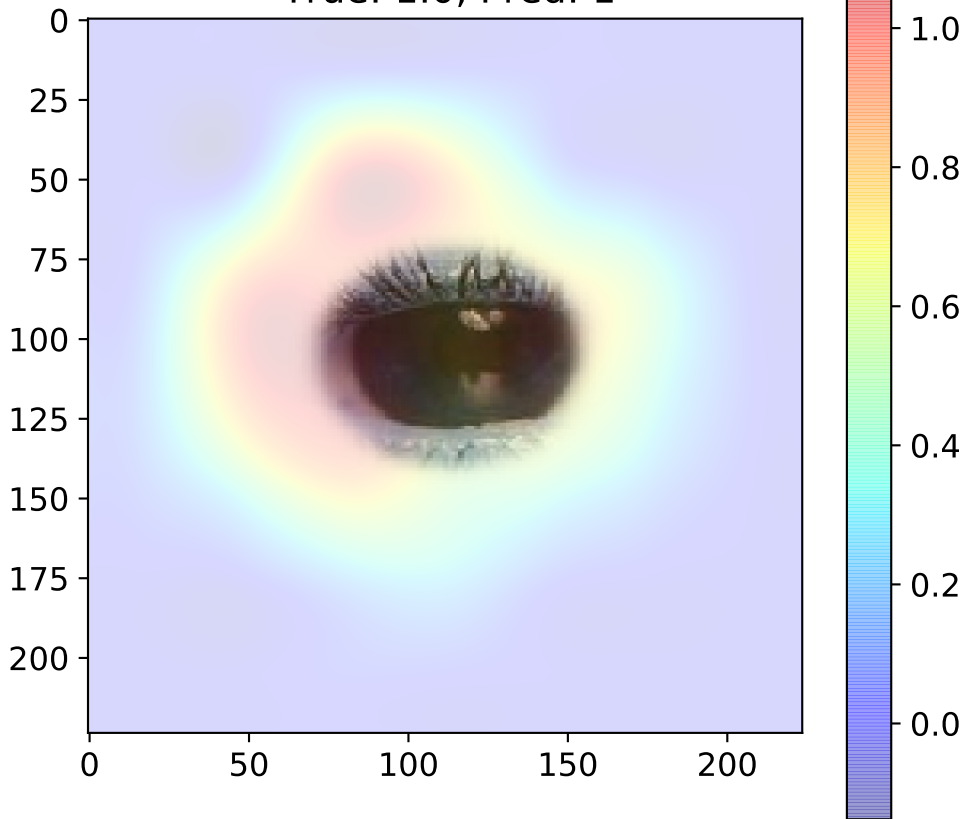
Cross-Validated Auc Over Epochs



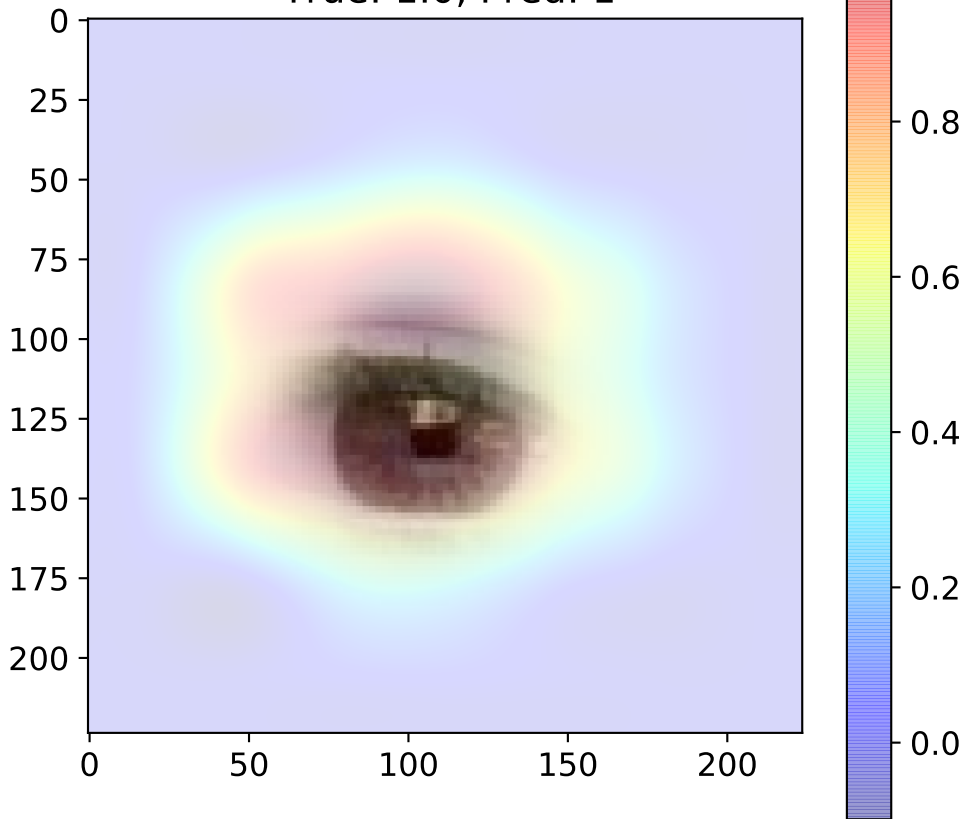
Cross-Validated ROC Curve - MobileNetV2



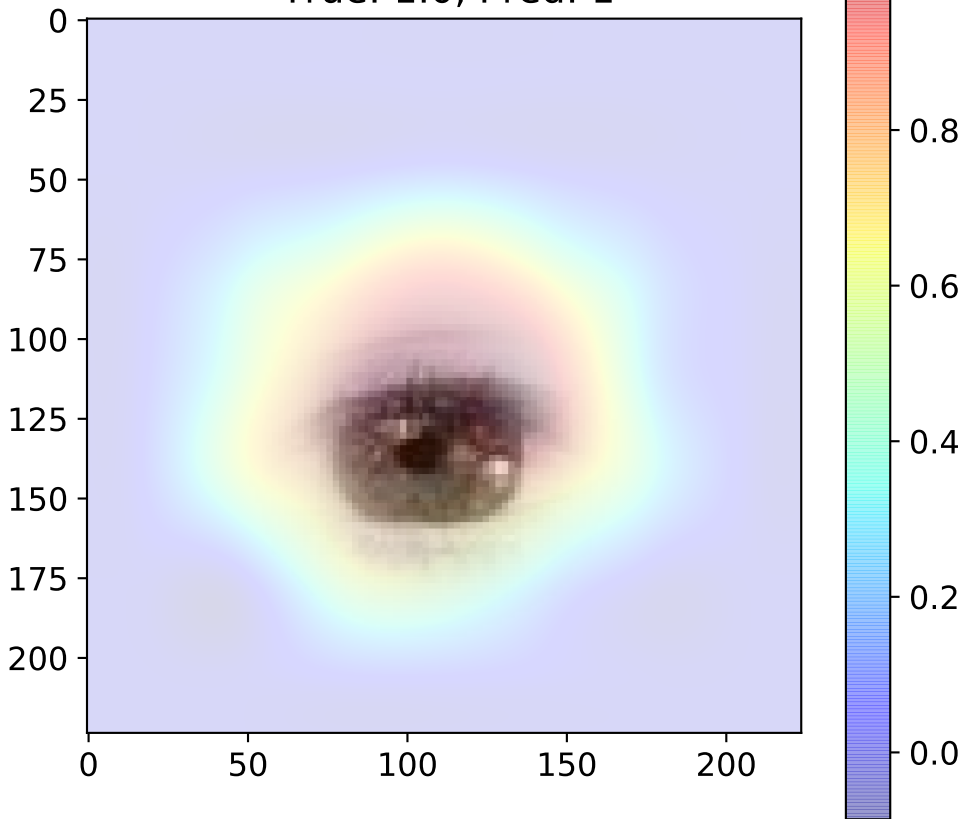
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



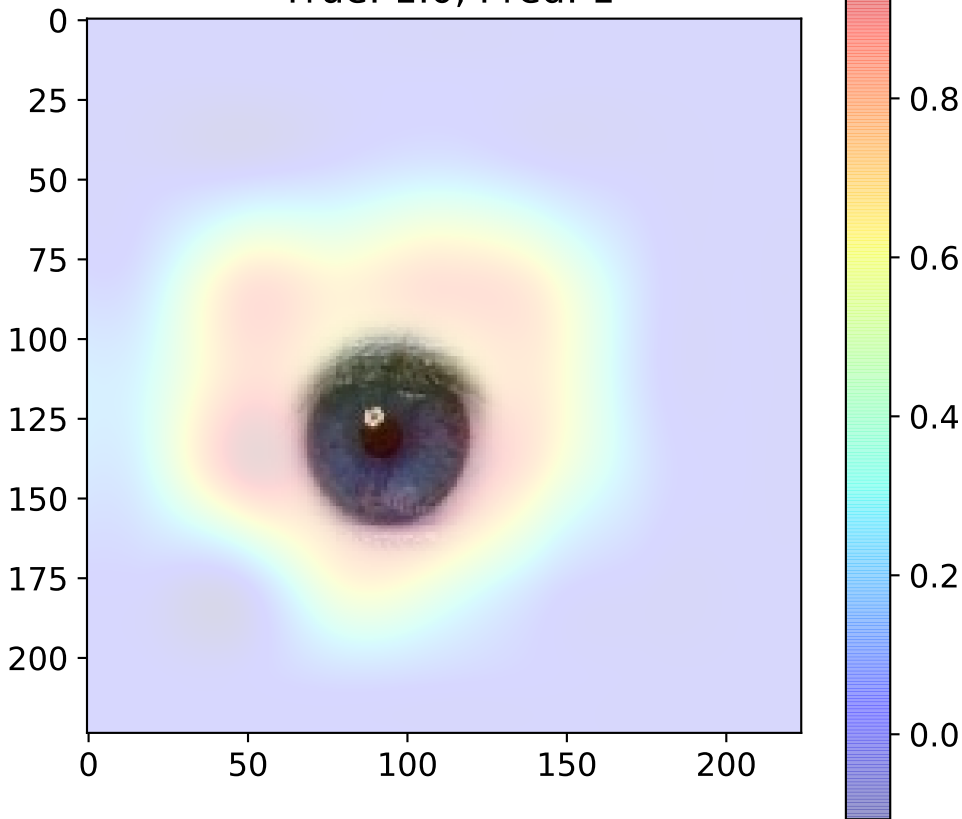
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



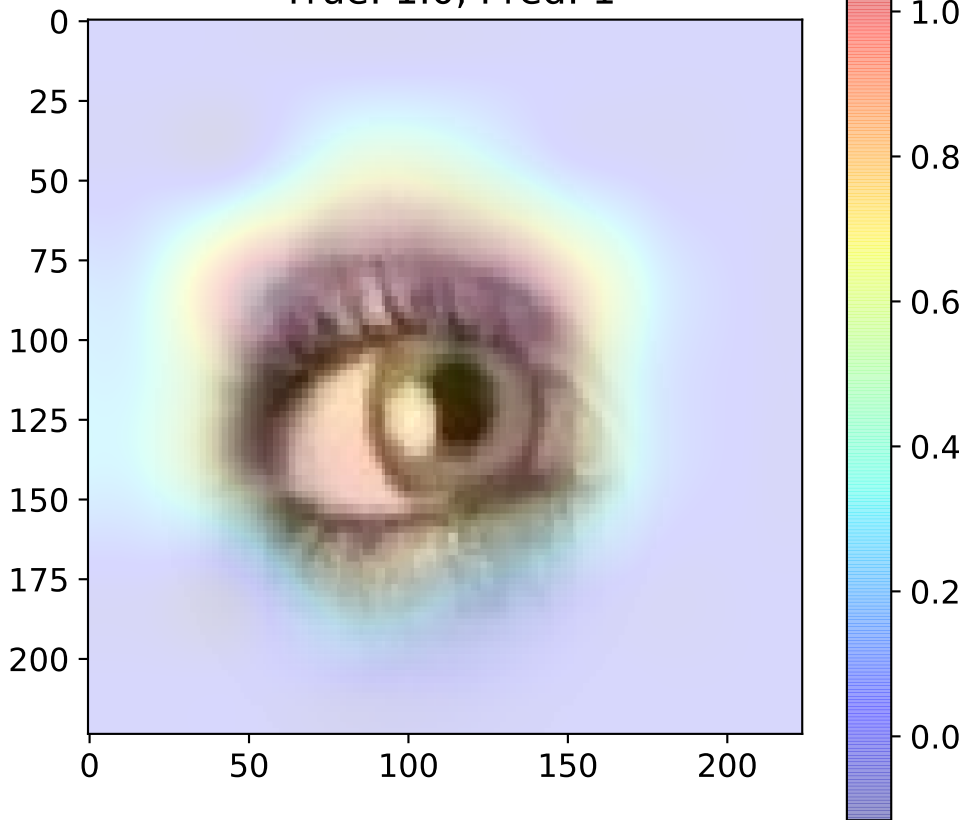
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



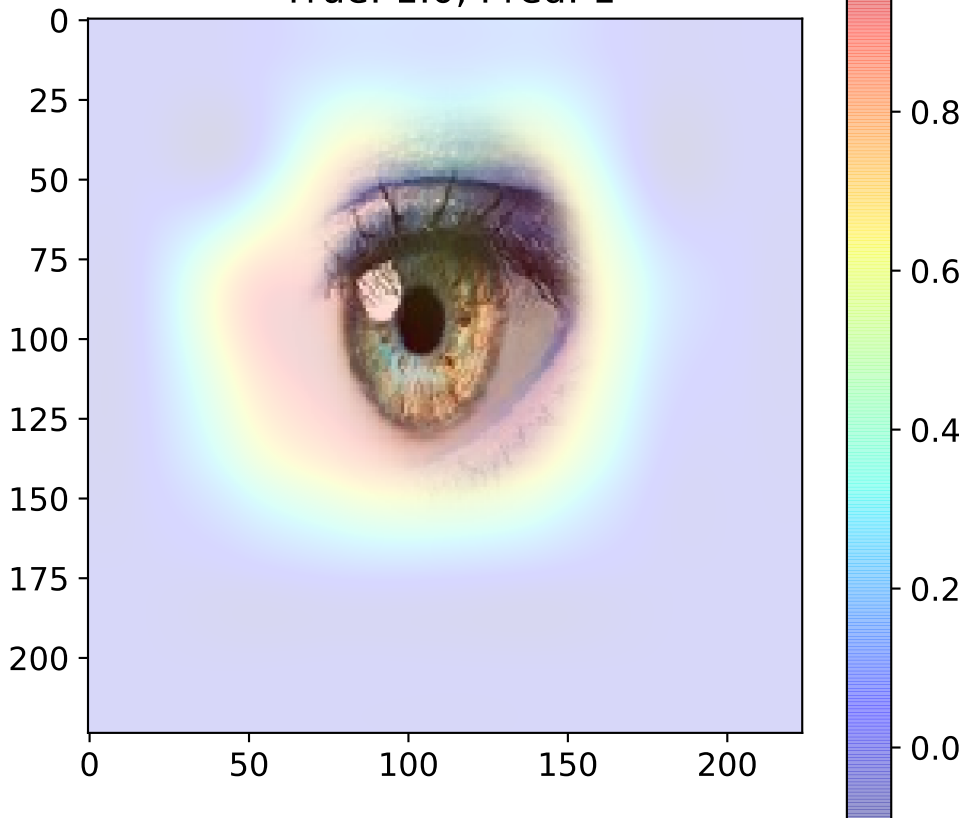
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



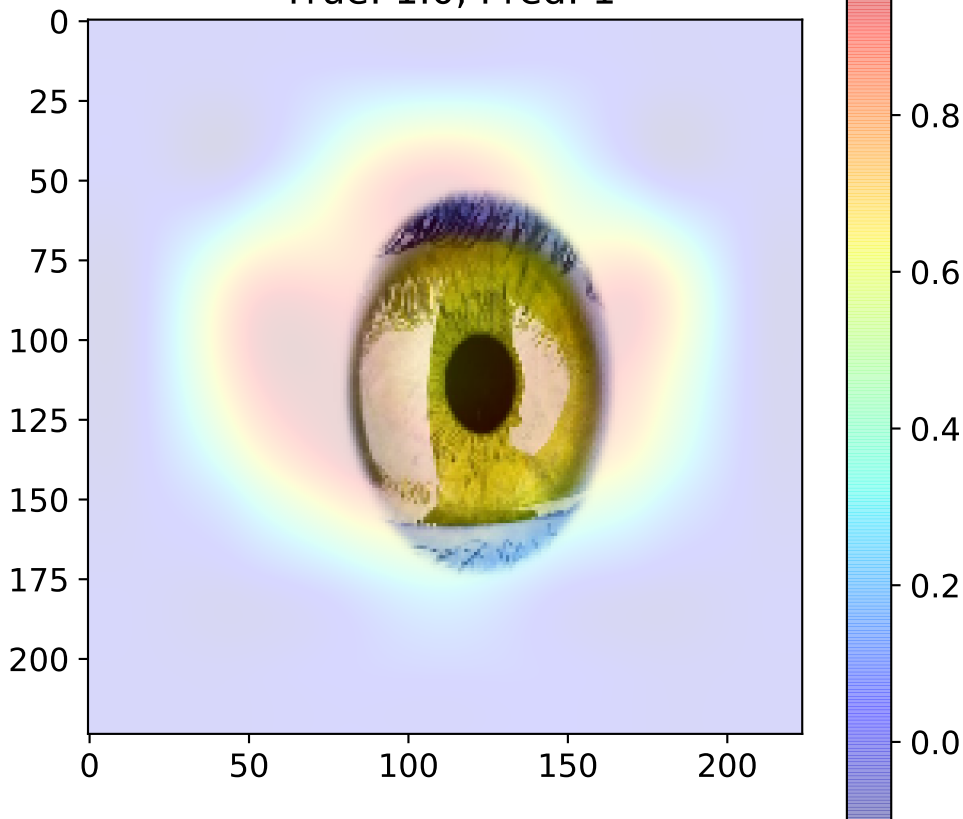
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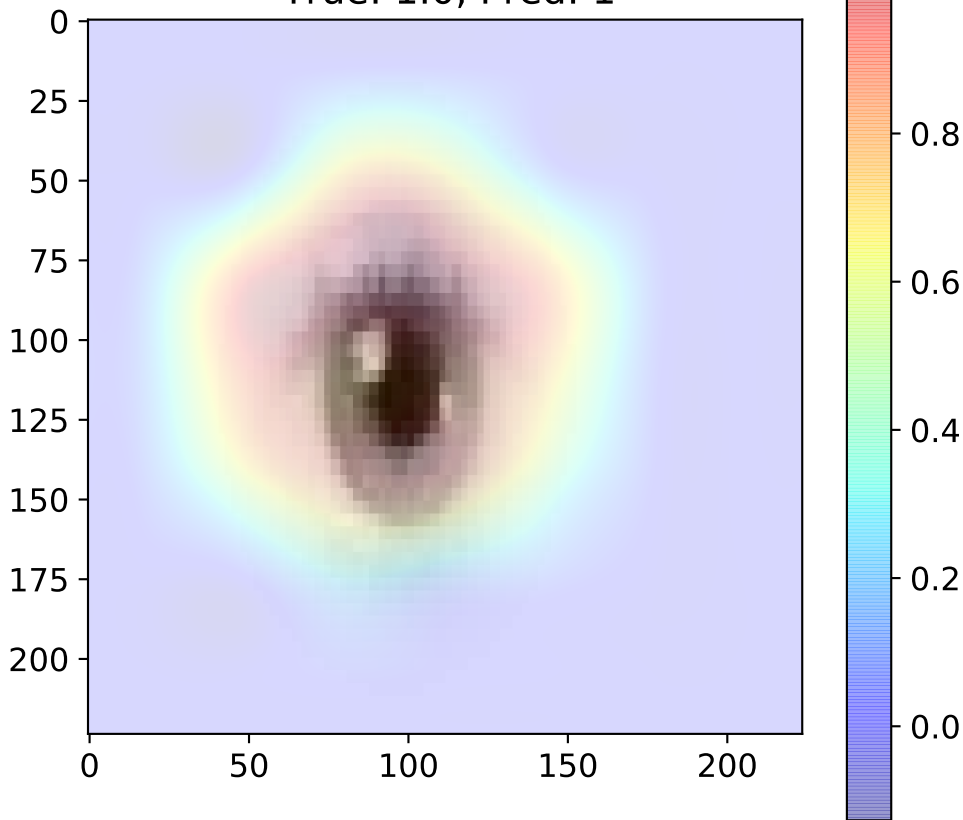


Correctly Classified (Best Fold)
True: 1.0, Pred: 1

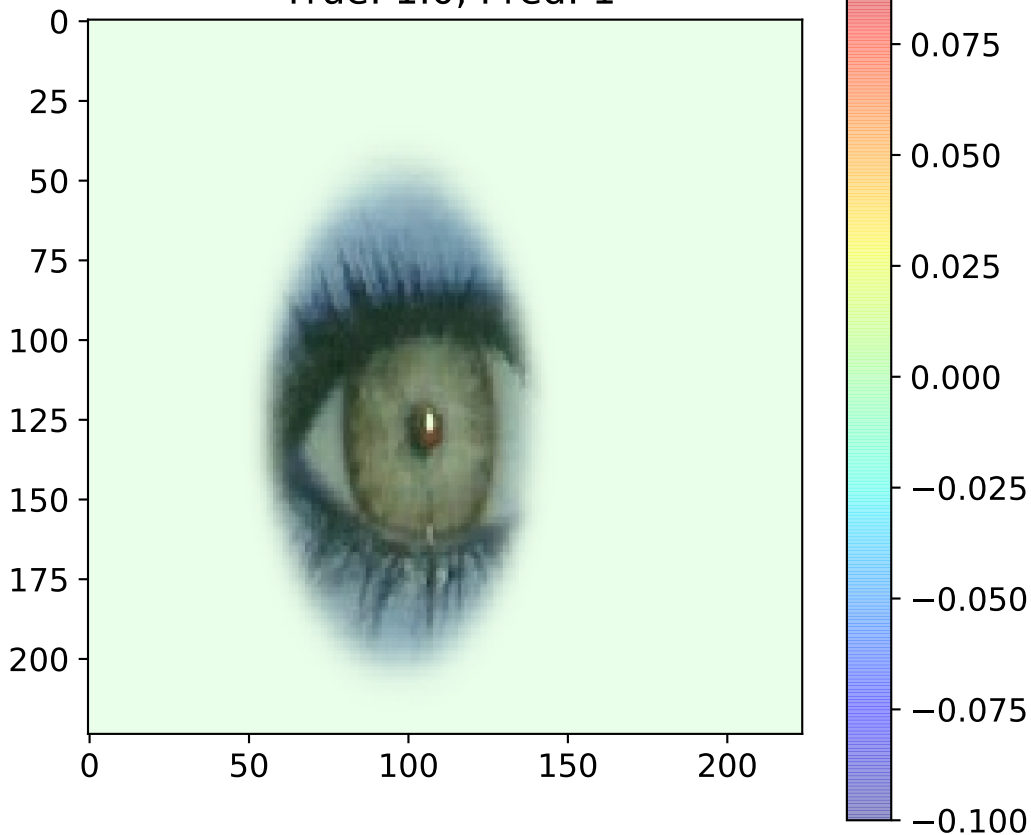


Correctly Classified (Best Fold)

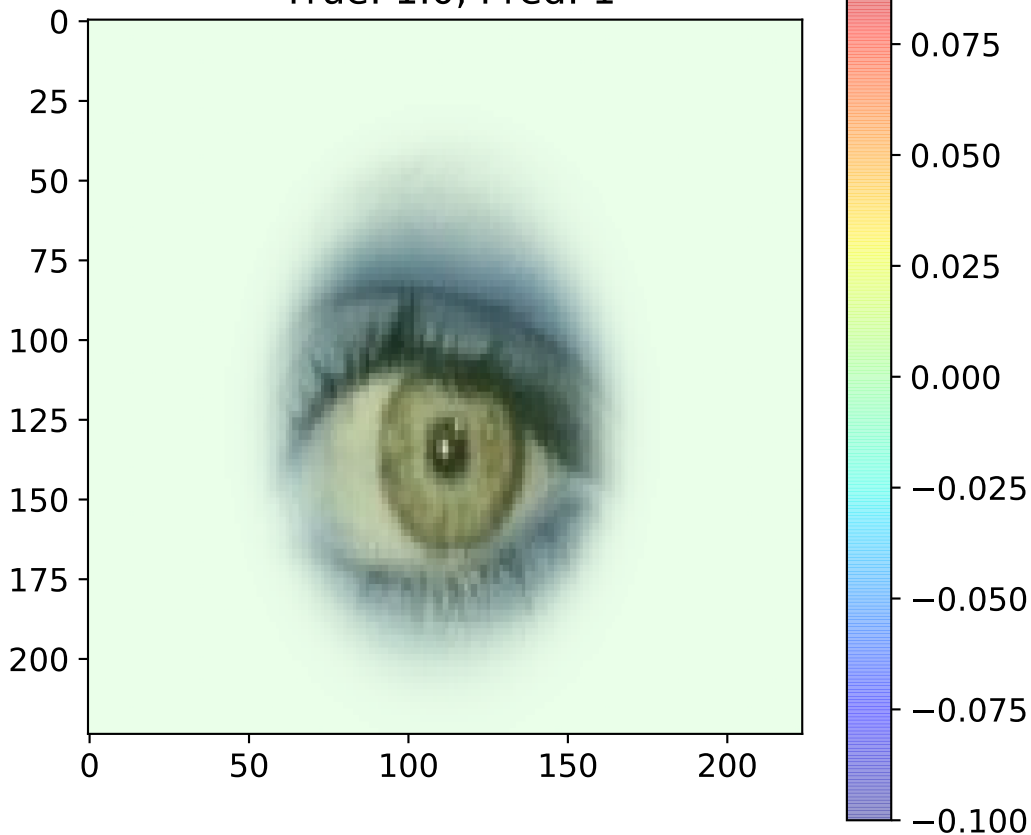
True: 1.0, Pred: 1



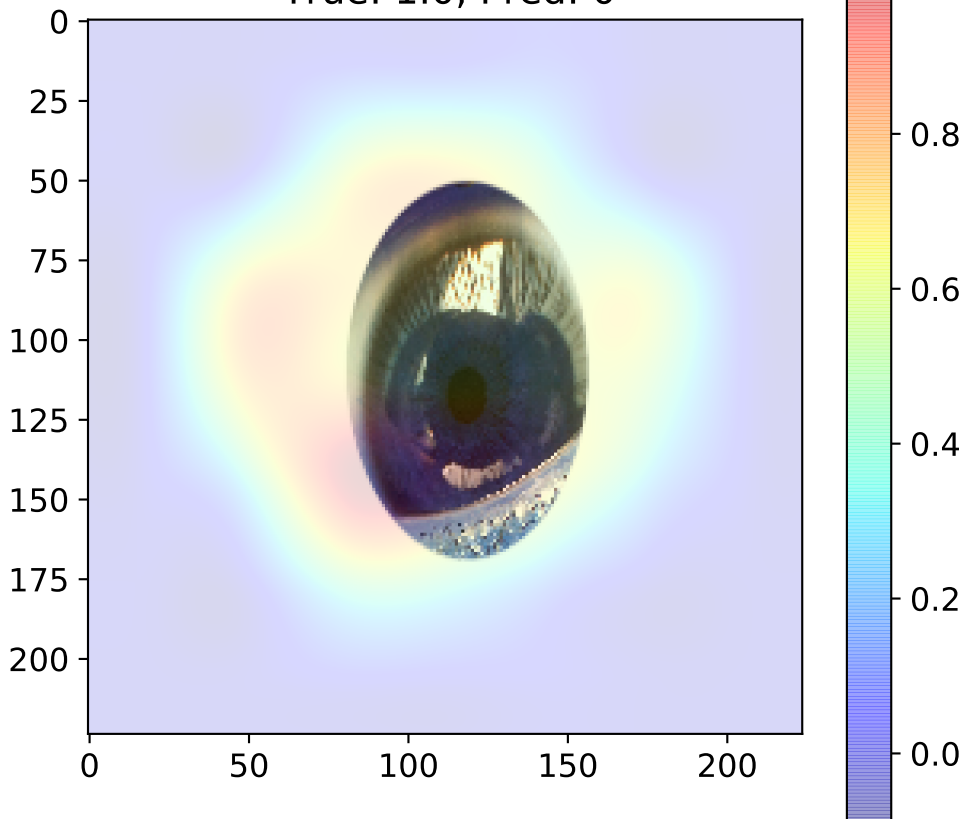
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



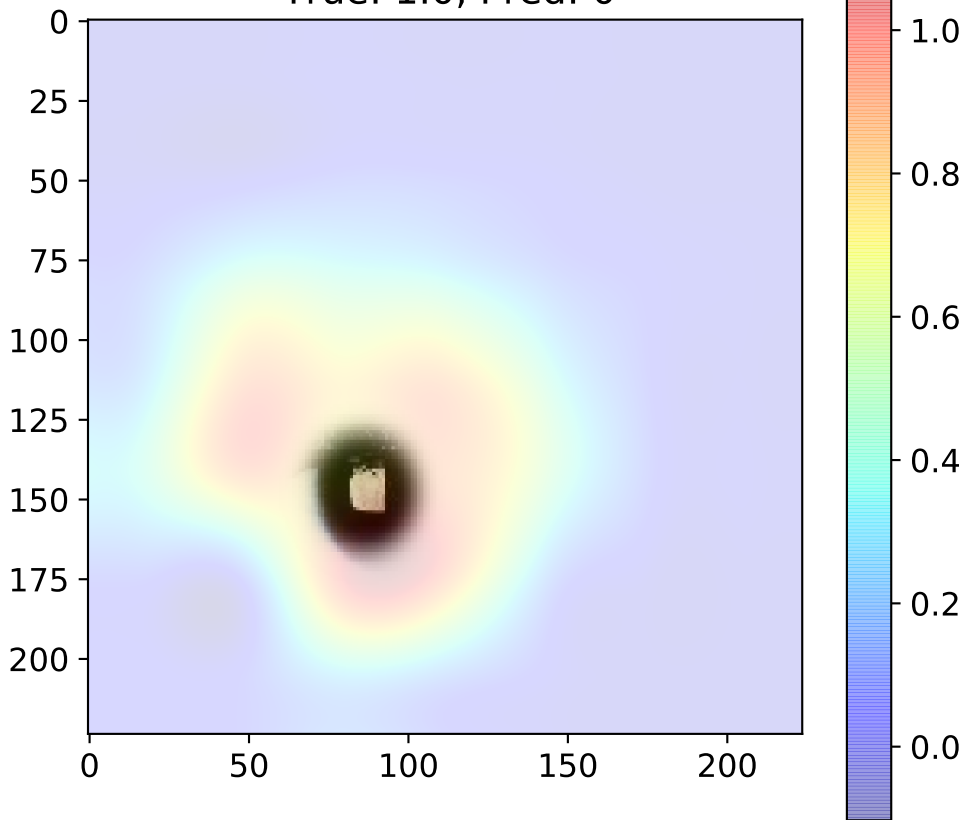
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



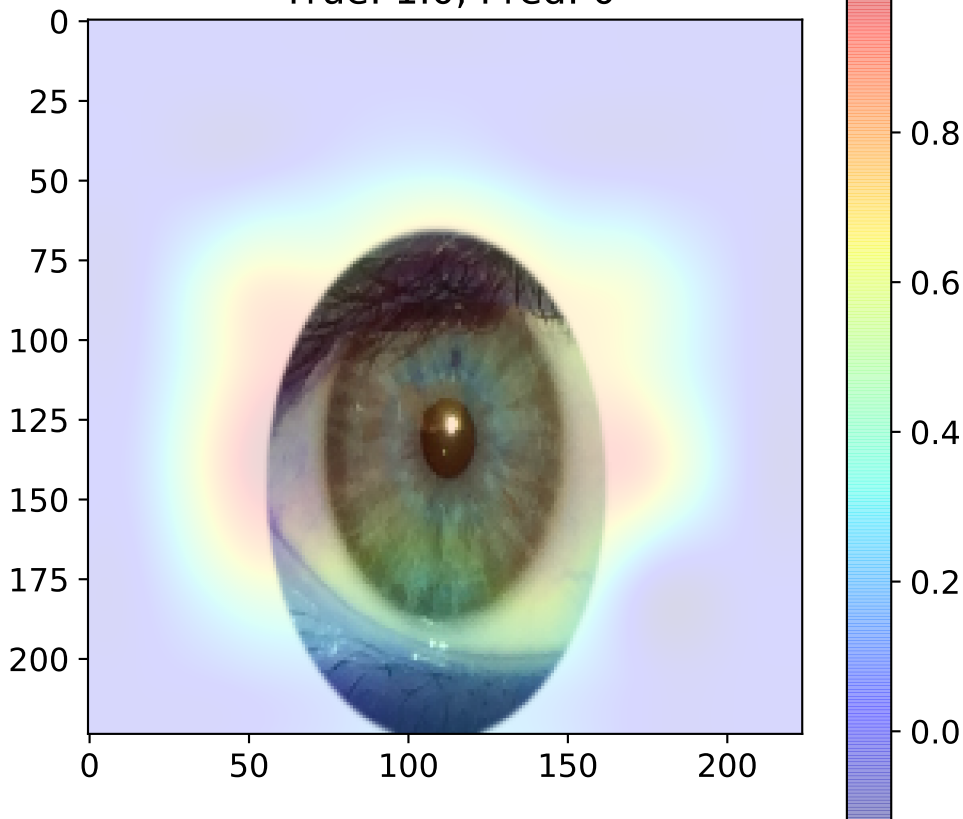
Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



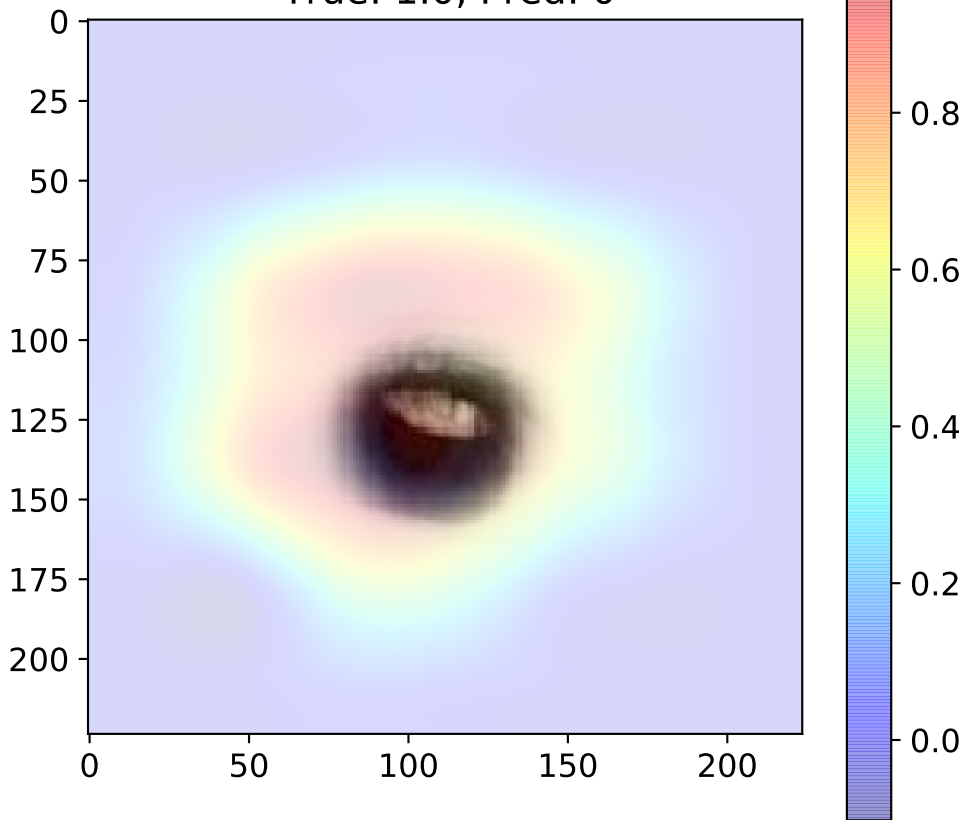
Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



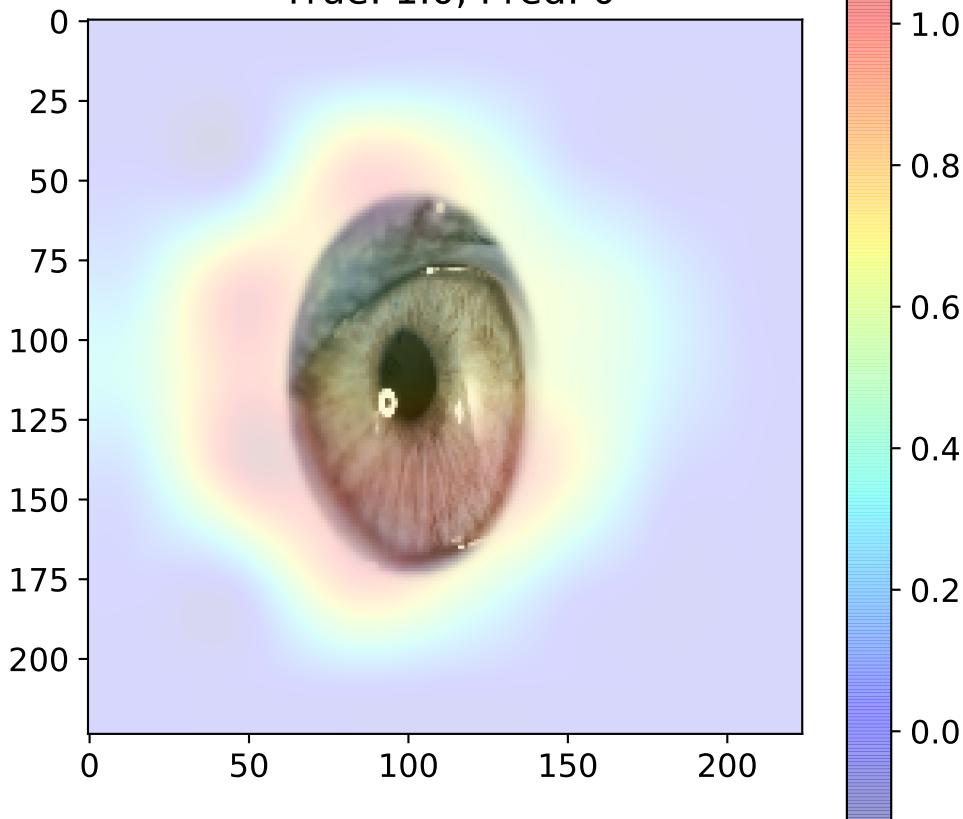
Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



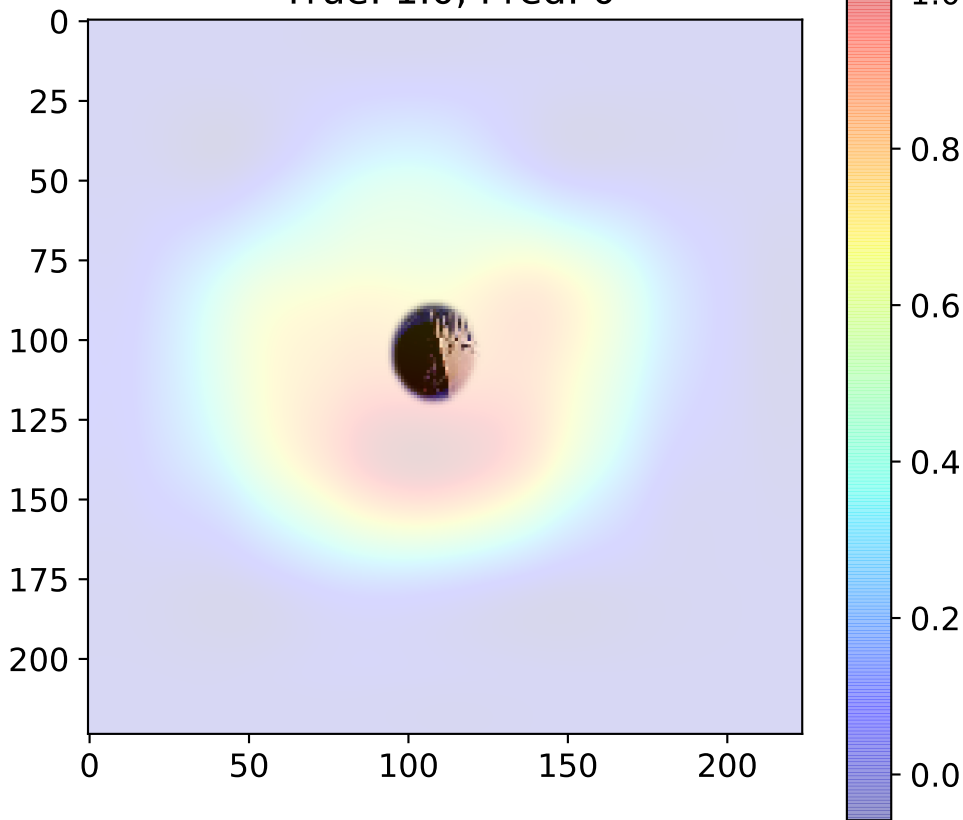
Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



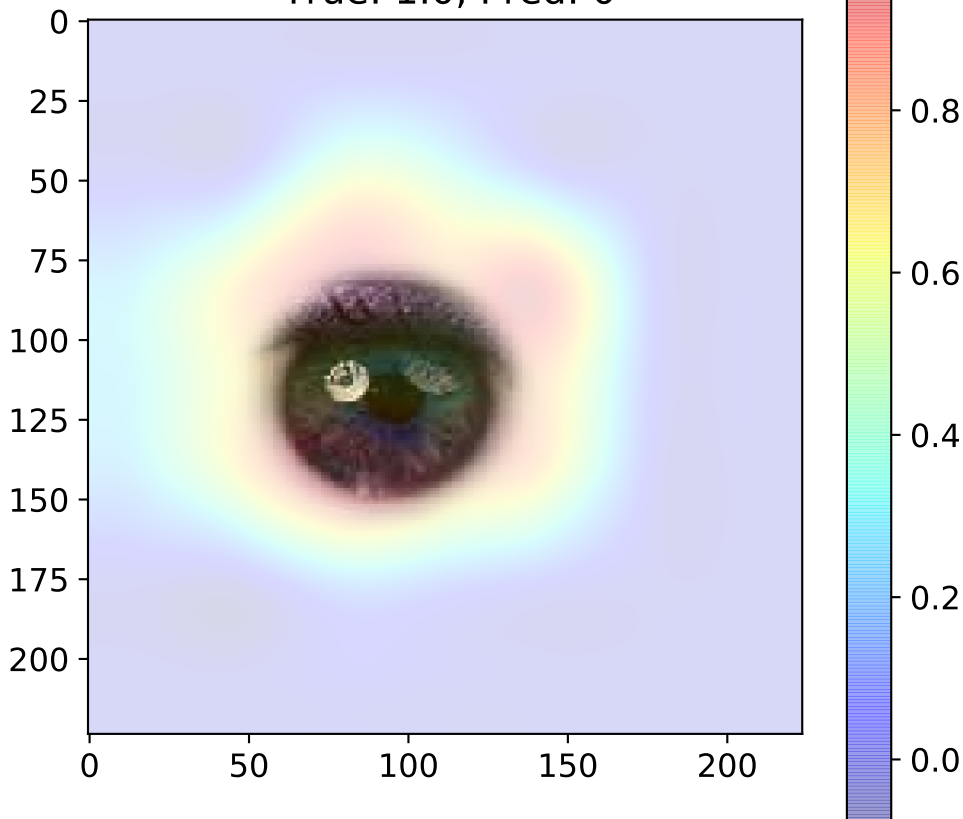
Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



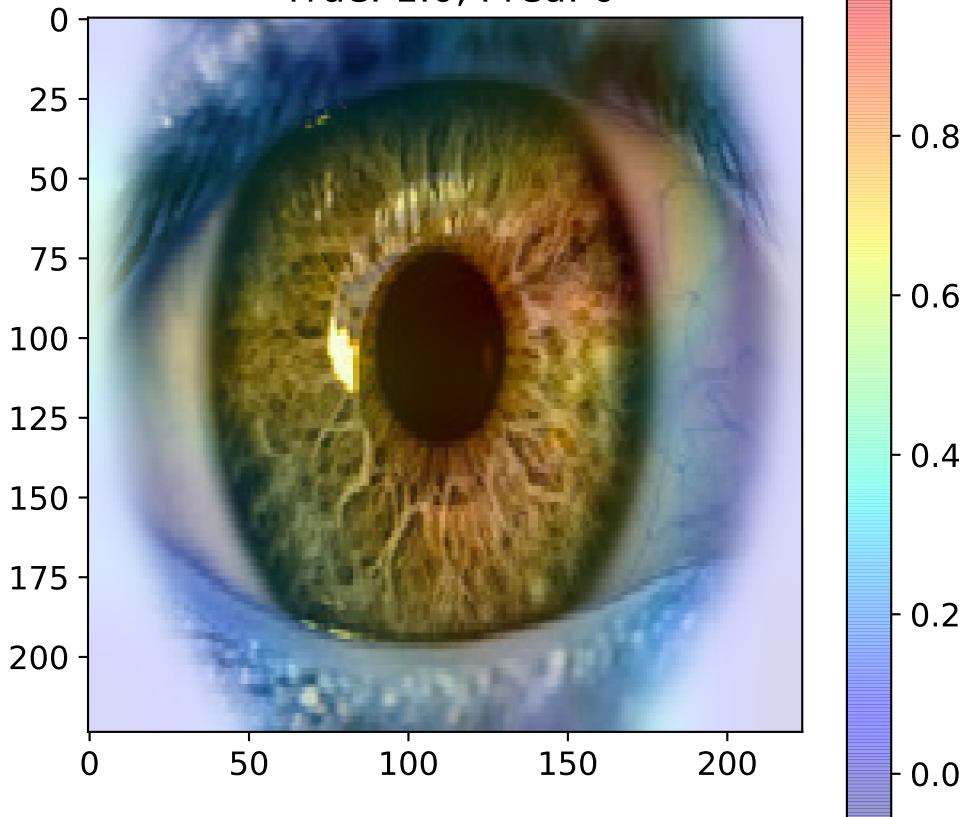
Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



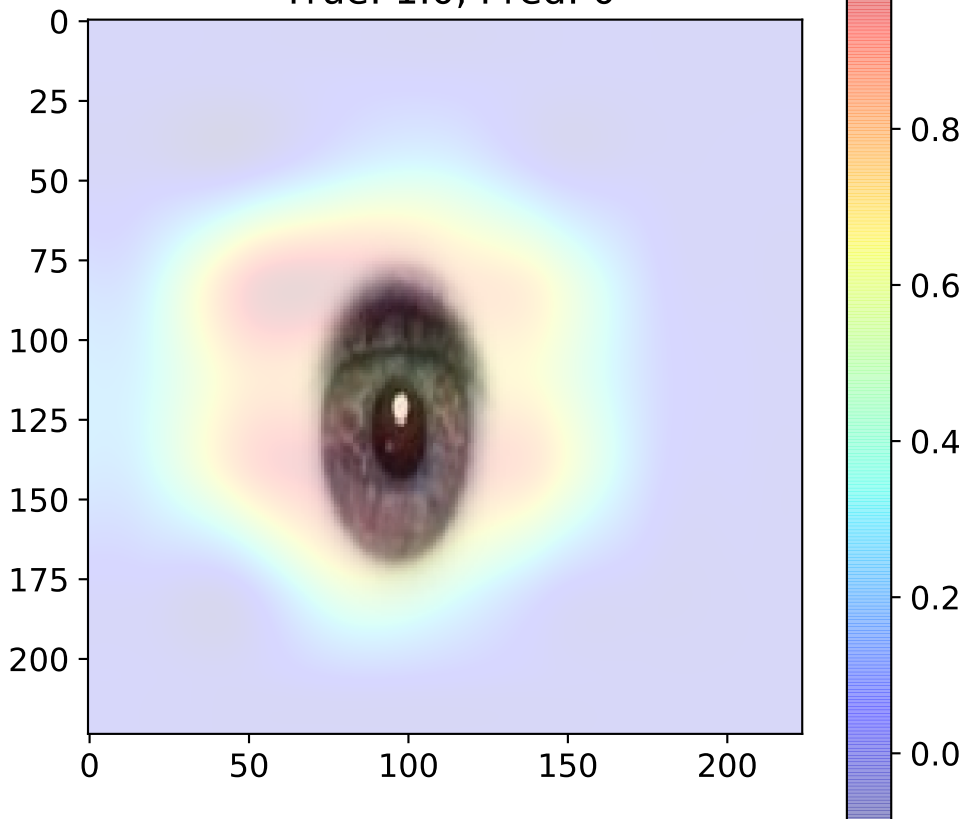
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Incorrectly Classified (Best Fold)

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